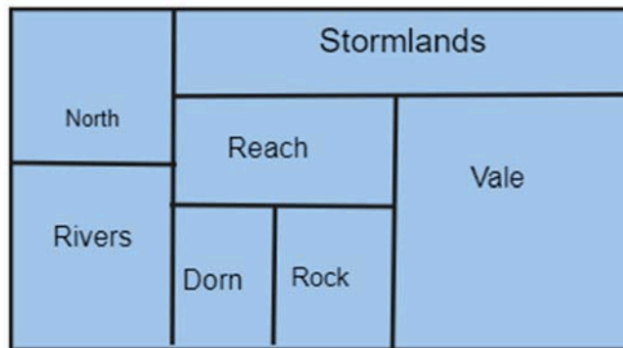


Practice problem (CSP)

Consider the following map of a Country where borders among the seven states have been marked. You need to color each of these divisions using three colors (Red/Green/Blue) such that two adjacent states do not receive the same color.

Now formulate the problem as CSP, Show the Constraint Graph, and solve the problem must applying both Minimum Remaining Values(MRV) and Least Constraining Value(LCV) heuristics. [2.5+2.5]



6. Consider the following map of Bangladesh where borders among the eight divisions have been marked. You need to color each of these divisions using three colors (Red/Green/Blue) such that two adjacent divisions do not receive the same color.

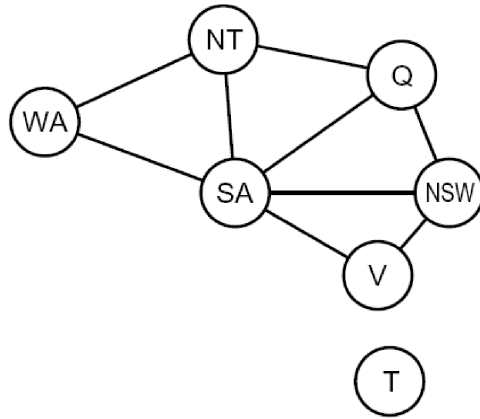
Now **formulate** the problem as CSP, Show the **Constraint Graph** and solve the problem **must applying** both Minimum Remaining Values(MRV) and Least Constraining Value(LCV) heuristics. [5]



For the given graph you have color all the nodes with red/green/blue so that the adjacent nodes don't contain same color:

i) Define the variables and domain for this problem

ii) Solve the problem as a CSP following **Minimum Remaining Value (MRV)** and **Least Constraining Value (LCV)** heuristics.



Sample Solution

i) Variables: WA, NT, Q, NSW, V, SA, T

Domain: {red, green, blue}

Step 1: Using MRV- all nodes have same number of remaining value (red, green value)

Break tie using degree heuristic, and choose SA.

Using LCV- all colors have the same impact, so choosing anyone (red).

SA←Red

Step 2: Using MRV- (WA, NT, Q, NSW, V) all have same number of remaining value

(green, blue)

Break tie using degree heuristics, and then NT, Q, NSW have the same number of edges.

So, we can choose any, now choosing NT.

Using LCV- all colors have same impact, so choosing anyone (green)

NT←green

Step 3: Using MRV- (WA,Q) both have one color remaining.

Break the tie with using degree heuristic, and choose Q.

Q←blue (only blue is the option)

Step 4: Using MRV- (WA,NSW) both have one color remaining.

Break the tie with using degree heuristic, and choose NSW.

NSW←green (only green is the option)

Step 5: Using MRV- (WA,V) both have one color remaining.

The both have same degree (same number of edge), so choose any.

WA←B

V←B

Step 6: T is independent (no edge) so put any color

T←red

6. A team of six employees is planning the seating arrangement for a new office layout. They are to be seated in any of the four departments – Finance, Marketing, HR, or IT. The following constraints apply:

- Employee A and Employee D want to be in the same department.
- Employee C prefers to work in the Finance department.
- Employees B and E are not on good terms and want to work in different departments.
- Employee D dislikes the Marketing department and will not sit there.
- Employee E is open to working only in HR or Finance
- Employee F must work in the same department as Employee C but in a different department from Employee E.

Formulate the problem as a CSP, clearly defining the variables, domains, and constraints, and solve the problem by applying both **Minimum Remaining Values(MRV)** and **Least Constraining Value(LCV)** heuristics. [6]

1. Four friends, Ryomen (R), Satoru (S), Yuji (Y), and Mahito (M) have come to a restaurant near their university. The restaurant serves Chicken Burger (C), Beef Burger (B), Kashmiri Naan (K), and Paratha (P). Their preferences are as follows:

- Ryomen does not like Paratha.
- Satoru and Yuji would like to grab a bite of each other's food, so they will order different dishes.
- Yuji likes burgers, so he will order a Chicken Burger or a Beef Burger.
- Ryomen likes to copy Satoru, so he will order the same dish as Satoru.
- Mahito will not order Kashmiri Naan because he had them earlier.
- Mahito is obsessed with being unique, so he will not order the same dish as anyone else.

Based on these constraints, answer the following questions:

- Draw the constraint graph for this CSP. [2]
- Solve the CSP using **MRV and LCV**. In case of any ties, choose the **alphabetically smaller** option. [6]

Practice problem (probability)

To understand which games are popular now, an online survey was conducted among 300 gamers about their favorite games. Among them, Valorant was the favorite game of 80 gamers among whom 50 were male. Marvel Rivals was the favorite of 60 gamers, 60% of whom were male. 70 gamers' favorite game was Overwatch, with 10% of them being female. The rest of the gamers loved Counter-Strike, and 10 were female

- a. Find the joint distribution table $P(\text{Game}, \text{Gender})$.
- b. Find $P(\text{Gender} \mid \text{Game}=\text{Valorant})$.
- c. If you pick a random male, what are the chances that he plays Marvel Rivals?

2. A survey was conducted on **300** Pokémon fans to analyze their preferences regarding Pokémon types. Half of the participants were **adults**, and the other half were **young** fans. Among the adults, the ratio of males to females is **2:1**, whereas among the young fans, **50%** are males and the rest are females. Among male adults, the preference distribution among **Fire-type**, **Water-type**, and **Grass-type** Pokémon follows the ratio **4:4:2**. Among female adults, the preference ratio is **3:3:4**. Among young male fans, the type preference is **2:2:1**, and among young female fans, the preferences for Fire, Water, and Grass types are equally distributed.

- a. Based on this data, construct the **full joint distribution table** over the three random variables: Category (C) $\in \{\text{Adult}, \text{Young}\}$, Gender (G) $\in \{\text{Male}, \text{Female}\}$ and Type (T) $\in \{\text{Fire}, \text{Water}, \text{Grass}\}$ [3]
- b. Find the **probability** that a randomly selected fan is a young male who prefers Fire-type Pokémon. [2]
- c. If a fan is randomly picked, and found to be a female young fan, what is the **probability** that she loves Water-types. [3]

d. Find the probability that a randomly selected fan is a young male and Water type pokemon fan.

e. Find the probability that a randomly selected fan is a young male or an Water type pokemon fan.

1. A survey was taken among **60 Senior** chess players and **40 Young** chess players. They were asked which **Chess pieces** they liked the most. Based on the data, the following **CPT (Conditional Probability Table)** was found. [Note: “**a**” and “**b**” are two variables that have exact values]

Age (A)	Chess Piece (C)	$P(C A)$
Young	Queen	a
Young	Knight	3a
Senior	Queen	2b
Senior	Knight	3b

- a) Find out the values of **a** and **b**. **[3]**
- b) What is the **percentage** of **Young** chess players who like the **Knight**? **[3]**

2. Suppose meningitis causes a patient to have a stiff neck 50% of the time. Moreover, the probability that any person has meningitis is 1 in 10,000 and the probability that any person without meningitis has a stiff neck is 5%. If a person has a stiff neck, what is the probability that he has meningitis?

Practice problem (Bayes' Net)

2. (a) A hospital is interested in modeling the probability of a patient having a disease (D). The disease causes two symptoms: fever (F) and fatigue (T). A positive blood test (B) is highly correlated with the presence of the disease, but not directly with the symptoms. Fatigue (T) can also be caused by lack of sleep (S), independent of the disease.

Each random variable has two possible events:

- Disease (D): {Yes, No}
- Fever (F): {Yes, No}
- Fatigue (T): {Yes, No}
- Blood test (B): {Positive, Negative}
- Sleep (S): {Sufficient, Insufficient}

(i) **Construct a Bayes net** based on this description, clearly indicating the relationships between the variables D (Disease), F (Fever), T (Fatigue), B (Blood test), and S (Sleep). [4]

(ii) Compare the memory complexity of storing the full joint probability distribution table with the memory required to store the Bayes net's conditional probability tables (CPTs). **Calculate the number of rows needed in each case.** [2]

3. (a) In a peer-reviewed academic conference, the Paper Acceptance decision is influenced by two main factors: **Reviewer Scores and Reputation of the Authors.**

- A **Well-Written Paper** increases the likelihood of high Reviewer Scores.
- Reviewer Scores also depend on the **Novelty of Findings**.
- A **Strong Publication History** increases the Author Reputation.

If a paper is accepted:

- It is more likely to be **Cited Early** after publication.
- The Author is **invited as a Speaker** at the next event.

- I. Construct the Bayes' Net from the above description. [2]
- II. Compare the memory complexity of storing the entries for Full Joint Distribution and the CPT of Bayes' Net. [2]

(b) Consider the following Bayes net:

- A (Rain) has two possible values: {True, False}.
- B (Sprinkler) has two possible values: {True, False}.
- C (Wet Grass) has two possible values: {True, False}.

The conditional probability tables (CPTs) are provided as follows:

Table 1: P(A)	
A	P(A)
True	0.3
False	0.7

Table 2: P(B A)		
A	B	P(B A)
True	True	0.4
True	False	0.6
False	True	0.6
False	False	0.4

Table 3: P(C A, B)			
A	B	C	P(C A, B)
True	True	True	0.9
True	True	False	0.1
True	False	True	0.8
True	False	False	0.2
False	True	True	0.7
False	True	False	0.3
False	False	True	0.1
False	False	False	0.9

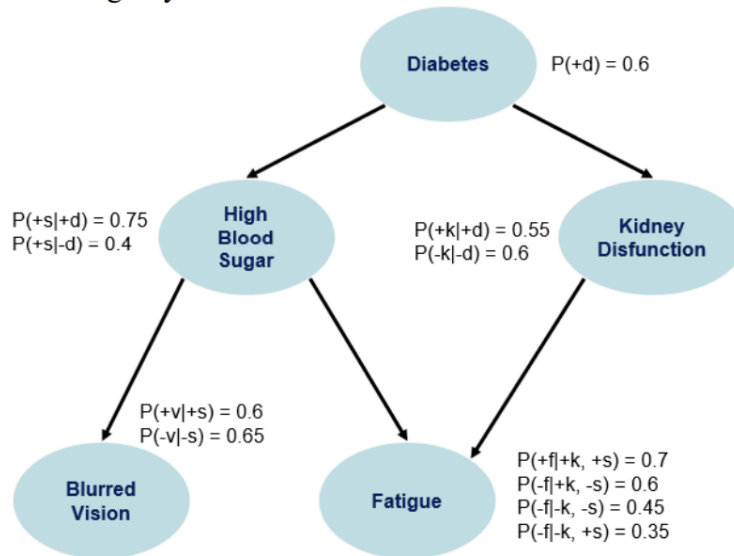
(i) Compute the joint probability $P(A = \text{True}, B = \text{True}, C = \text{True})$.

[3]

(ii) Compute $P(A = \text{True} \mid C = \text{True})$ using Bayes' rule.

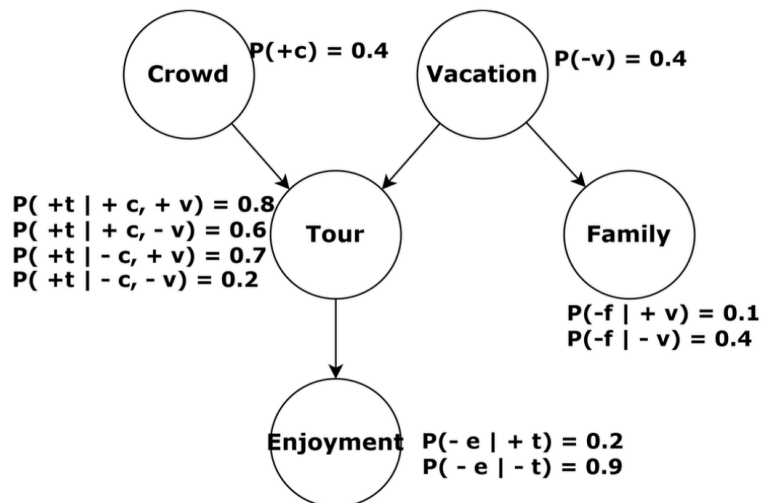
[3]

(b) Consider the following Bayes' Net:



Find the probability of a person having fatigue given he has diabetes using Inference by Enumeration. [4]

c) Given a new Bayesian network below, find out the Probability of Enjoyment (+e) given there is no crowd (-c). [5]



Practice problem (Naive Bayes)

Dr. Eliza Thornberry is researching animal behaviors in various environments. The following dataset logs observed features and the outcome behavior of animals in each observation. Use a Naive Bayes Classifier with **Laplace smoothing (k = 2)** to determine the most likely behavior outcome for a random animal event with attributes: <Evening, Mountain, Heavy Rain, Noisy> [8]

Obs No	Time	Terrain	Weather	Sound	Behavior
1	Morning	Forest	Sunny	Quiet	Playful
2	Afternoon	Mountain	Sunny	Quiet	Playful
3	Morning	Desert	Windy	Noisy	Sleepy
4	Evening	Forest	Heavy Rain	Noisy	Playful
5	Night	Desert	Windy	Quiet	Calm
6	Evening	Mountain	Snowy	Noisy	Calm
7	Afternoon	Forest	Windy	Quiet	Sleepy
8	Evening	Mountain	Heavy Rain	Quiet	Playful
9	Morning	Mountain	Snowy	Quiet	Calm
10	Night	Forest	Heavy Rain	Noisy	Sleepy
11	Afternoon	Desert	Sunny	Quiet	Playful
12	Night	Forest	Snowy	Quiet	Calm
13	Morning	Desert	Sunny	Noisy	Playful
14	Afternoon	Mountain	Windy	Quiet	Playful
15	Evening	Forest	Sunny	Noisy	Calm

4. Consider a set of documents, each of which either categorized as *Sports* or *Informatics* category. These documents are categorized based on the presence of 8 words within that document: { **goal, tutor, variance, speed, drink, defence, performance, field** }

Training data set:

goal	Tutor	variance	speed	drink	defence	performance	field	category
Yes	No	No	No	Yes	Yes	Yes	Yes	<i>Sports</i>
No	No	Yes	No	Yes	Yes	No	No	<i>Sports</i>
No	Yes	No	Yes	No	Yes	Yes	No	<i>Sports</i>
No	No	Yes	No	Yes	No	Yes	No	<i>Informatics</i>
No	No	No	No	No	No	No	No	<i>Informatics</i>
No	No	Yes	Yes	No	No	Yes	Yes	<i>Sports</i>
No	Yes	Yes	No	No	Yes	No	No	<i>Informatics</i>
Yes	No	No	No	Yes	No	Yes	Yes	<i>Sports</i>
Yes	Yes	No	Yes	No	No	Yes	Yes	<i>Informatics</i>
No	Yes	Yes	No	No	No	Yes	No	<i>Informatics</i>
Yes	No	No	Yes	No	yes	No	Yes	<i>Sports</i>

- a. Now predict the category of a new document containing the following words: [goal=Yes, tutor=No, variance=No, speed=Yes, drink=Yes, defence=No, performance=Yes, field=Yes] **[5]**
- b. Briefly explain why we need Laplacian Smoothing during Naïve Bayes Classification. **[2]**

6. Suppose the following data is collected from an eye hospital on what kind of contact lenses are suitable for a patient:

Age	vision	astigmatism	Tear production	Lens type
Young	Near	No	Reduced	None
Young	Near	No	Normal	Soft
Young	Near	Yes	Reduced	None
Young	Far	No	Normal	Soft
Young	Far	Yes	Normal	Hard
Middle-aged	Near	No	Reduced	None
Middle-aged	Near	No	Normal	Soft
Middle-aged	Near	Yes	Normal	Hard
Middle-aged	Far	Yes	Reduced	None
Middle-aged	Far	No	Normal	Soft
Old	Near	Yes	Normal	Hard
Old	Near	No	Reduced	None
Old	Far	No	Reduced	None
Old	Far	No	Normal	Soft
Old	Far	Yes	Reduced	None

Given that a patient is old, has far sightedness, has astigmatism and tear production is normal, what lens type should be suggested? Determine using Naïve Bayes classifier with LaPlacian smoothing constant $k=1$. [5]

Practice problem (Decision Tree)

4. You are given the opportunity to build a decision tree to predict the **performance rating** of employees based on the given attributes. Using the data below, determine which attribute is the **most important factor** in evaluating an employee's performance. [8]

Experience Level	Teamwork	Project Delivery	Performance Rating
High	Excellent	On Time	Excellent
Medium	Good	On Time	Good
Low	Poor	Late	Poor
High	Good	On Time	Excellent
High	Excellent	Late	Good
Medium	Poor	Late	Poor
Medium	Good	On Time	Good
Low	Good	Late	Poor
Medium	Excellent	On Time	Excellent
Low	Poor	Late	Poor
High	Good	On Time	Excellent
Medium	Good	On Time	Good

4. In a retail system, we are analyzing customer purchasing behavior based on three attributes: **Product Category, Price Range, and Customer Rating**. The goal is to determine how these attributes influence the decision to buy a product (Yes/No).

Product Category	Price Range	Customer Rating	Buy
Electronics	High	5	Yes
Fashion	Low	4	No
Electronics	Low	5	Yes
Home & Kitchen	Low	3	No
Fashion	High	4	Yes
Electronics	High	3	No
Fashion	Low	5	Yes
Home & Kitchen	Low	4	No
Electronics	High	4	Yes
Fashion	High	3	No
Home & Kitchen	Low	5	Yes
Electronics	Low	3	No
Fashion	High	4	Yes
Home & Kitchen	High	3	No
Electronics	Low	5	Yes

Your task is to learn a Decision tree based on this data to predict whether a particular customer will buy the product or not. **Which should be the attribute in the root node of the decision tree?** Just find the attribute at the root node. Show detailed calculations. **[8]**

Practice problem (Markov Model)

5. A music analytics company is studying listening patterns on **Spotify**, specifically tracking how users switch their primary artist preference among three major rap artists: **Eminem**, **Travis Scott**, and **Drake**. They collected data on users' most-played artists in **January 2023** and tracked how their preferences changed by **January 2024**.

- **Initial (January 2023)** survey results (**total of 180 users**): **80** primarily listened to **Eminem**, **60** listened to **Travis Scott**, **40** listened to **Drake**

One year later (**January 2024**), the following preference shifts were observed:

- Among the **80 Eminem** listeners in **2023**, **16** switched to **Travis Scott** as their primary artist, **12** switched to **Drake**, and the rest stayed loyal to **Eminem**
- Among the **60 Travis Scott** listeners in **2023**, **15** switched to **Eminem**, **9** switched to **Drake** and the rest continued with **Travis Scott** as their top artist
- Among the **40 Drake** listeners in **2023**, **10** switched to **Eminem**, **6** switched to **Travis Scott** and the rest kept **Drake** as their most-played artist

Assume these **transition patterns** remain the same for each subsequent year (**2024 → 2025**, **2025 → 2026**, **2026 → 2027**, etc.).

- a) Construct a **transition matrix** representing the listener preference changes from any year **t** to the next year **t+1**. [2]
- b) Predict the percentage of users who will have **Travis Scott** as their primary artist in **2028** using the initial survey of **2023**. [3]
- c) Calculate the **long-run probabilities** that a randomly selected Spotify user will have for each of the three artists. [3]